

Dobie Zweli Langeni

Johannesburg, Gauteng, South Africa

0713534898 | langenidzweli@gmail.com | [LinkedIn](#)

Professional Summary

Final-year BSc Mathematical and Computer Science student (Class of 2026) with strong foundations in algorithms, data structures, and software engineering, and hands-on experience building full-stack and backend systems using Java (Spring Boot), JavaScript, and MySQL. Developed real-world applications including a restaurant POS system and a secure student records platform, focusing on clean architecture, modular design, and scalability.

Demonstrated leadership as Deputy President of a student coding society, organizing workshops and mentoring peers in software development. Selected as a university tutor for 50+ first-year students in Object-Oriented Programming (Java), supporting the development of strong programming fundamentals and problem-solving skills. Passionate about solving real-world problems through technology and continuously improving as an engineer, with a strong interest in scalable backend systems and high-performance software development.

Experience

Student Tutor – Object-Oriented Programming (Java)

Sol Plaatje University

- Selected as a university tutor supporting 50+ first-year students across a full-semester Java module covering OOP, data structures, exception handling, file I/O, and GUI development
- Delivered structured tutorials on object-oriented design, debugging strategies, and event-driven programming to strengthen core programming fundamentals
- Provided one-on-one academic support, improving students' code correctness, debugging efficiency, and problem-solving ability
- Guided students in developing modular, maintainable applications, reinforcing clean code practices and software engineering principles

Deputy President

Circuit Breakers – Student Coding Society

- Led peer-driven programming sessions in Java and C++, supporting 100+ students in strengthening data structures knowledge and algorithmic problem-solving skills
- Designed and delivered hands-on workshops focused on building real-world applications, including full-stack and desktop systems
- Coached members in collaborative software development practices, including Git-based version control, modular system design, and structured problem-solving approaches
- Promoted a high-performance learning culture within the society by encouraging ownership, accountability, and continuous technical growth

Education

BSc Mathematical and Computer Science

Sol Plaatje University | 2024 – 2026 (Expected)

- Final-year student with strong foundations in software engineering, algorithms, and systems design, supported by a solid background in mathematics and statistics. Focused on applying structured problem-solving and computational thinking to design efficient, scalable software systems.
- Core focus areas:** Software Engineering, Algorithms & Data Structures, Operating Systems, Object-Oriented Programming, Statistics.

Skills

Technical Skills

- **Programming Languages:** Java, JavaScript, Python, C++, SQL/MySQL, HTML/CSS
- **Backend & Frameworks:** Spring Boot, RESTful API development, Java Swing
- **Core Software Engineering:** Object-Oriented Programming (OOP), Data Structures & Algorithms, System Design, Modular Software Design, Clean Code Principles
- **Tool & Platforms:** Git, VS Code, MySQL

Engineering & Professional Competencies

- Full-stack and backend system development using RESTful APIs and database-driven architectures
- Design and implementation of scalable, modular, and maintainable software systems
- Debugging, testing, and optimisation of Java-based applications
- UI development for desktop applications using Java Swing
- Strong problem-solving ability applying algorithms and computational thinking
- Leadership and mentoring in collaborative software development environments
- Effective communication and teamwork in project-based development settings

Projects

The Heights Restaurant POS | *Java (Spring Boot), HTML, CSS, JavaScript*

- Developed a full-stack restaurant management system supporting real-time table management, order processing, and payment tracking.
- Built a Spring Boot backend exposing RESTful APIs to manage order workflows, staff operations, and system communication between components.
- Designed a dynamic frontend to visualize table availability and active orders in real time, improving operational awareness and usability.
- Implemented structured backend logic using data models and algorithmic processing to optimize order flow and automate reporting.

GitHub: <https://github.com/Langenidzweli/The-Heights-Restaurant-POS>

Student Records Management System | *Node.js, Express.js, MySQL, React*

- Built a secure full-stack academic records system for managing student assessments, results, and performance tracking.
- Implemented role-based authentication and authorization using JWT to ensure secure access for lecturers and administrators.
- Developed a responsive React interface for academic data entry, result processing, and reporting workflows.
- Designed a relational MySQL database with constraints, indexing, and triggers to maintain data integrity and consistency.

GitHub: <https://github.com/Langenidzweli/Database-Student-Record-System>

Tip Calculator Desktop Application | *Java, Java Swing*

- Developed a multi-screen desktop application for tip calculation, bill splitting, and transaction history tracking.
- Applied object-oriented design principles to separate UI, business logic, and data layers, improving maintainability and scalability.
- Implemented event-driven UI interactions with validation logic to ensure accurate and reliable calculations.

GitHub: <https://github.com/Langenidzweli/Java-Swing-Tip-Calculator>

Virtual Guard (In Progress) | Software Engineering / System Design (Tech stack evolving)

- Designing a scalable security monitoring system focused on real-time event detection, tracking, and automated alerting to improve situational awareness and response efficiency.
- Developing backend system components responsible for processing incoming events, maintaining system state, and triggering automated responses within simulated real-world environments.
- Applying software engineering principles to design a modular, extensible architecture with emphasis on performance, reliability, and scalability.
- Exploring rule-based logic and decision-making mechanisms to support intelligent event classification and automated response behaviour.